

RUIYANG WANG

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Summary

Applied ML and Robotics researcher experienced in building scalable robot learning and planning systems with **PyTorch** and **ROS2**. Skilled in **multi-robot coordination**, **sensor fusion**, and **language conditioned reasoning**. Hands-on with **Dockerized simulation**, **modular pipeline design**, and **real-world robotic deployment**.

Skills

Programming & Tools: Python, C++, Julia, MATLAB, ROS2, Linux/Unix, Git, Docker

Tools & Cloud: AWS S3/EC2, MLflow, Weights & Biases, OpenAI API

ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn

Robotics & Simulation: ROS2, LCM, PyBullet, OpenAI Gym, Nvidia Isaac Lab

Education

Duke University - Durham, NC

August 2024 - April 2028 (Expected)

Doctor of Philosophy (Ph.D.): Electrical and Computer Engineering

University of Michigan, Ann Arbor - Ann Arbor, MI

August 2018 - December 2023

Master of Science: Robotics

Bachelor of Science: Mechanical Engineering

Applied ML/Robotics Projects

Large Language Model Multi-robot Coordinated Exploration and Search *January 2025 - September 2025*

PyTorch, ROS2, OpenAI API, Unitree Go2

- Built **LLM-MCoX**, a large-scale multi-robot coordination framework integrating LiDAR frontier clustering, doorway detection, and LLM reasoning.
- Enabled **semantic guidance** from natural-language priors (e.g., “object likely in northeast corridor”) to optimize exploration and search tasks.
- Validated in 100+ simulated and real-world trials with Unitree Go2 and X500 drone; achieved 22% faster exploration and **50% higher search efficiency** than baseline frontier-based methods. **[P1]**

Neural-Symbolic Deadlock Resolution in Multi-robot Systems

August 2024 - December 2024

PyTorch, PyBullet, Neural Logic Machines

- Built a **Neuro-Symbolic planner** combining Neural Logic Machines for post-deadlock resolution in multi-robot systems.
- Trained models on **small 2–5 robot scenarios**, generalizing to **10+ robot** configurations in cluttered layouts. Ensured post-occurrence deadlock resolution in multi-robot coordination, where existing methods often resulted in incomplete task execution. **[P2]**

* both authors have equal contribution

Compatible Constraint Selection in Receding-Horizon Control *Optimization, MPC, MATLAB, Python*

December 2022 - December 2023

- Implemented heuristics for **maximal feasible subset selection** in nonlinear MPC using Lagrange multipliers to identify conflicting soft constraints.
- Achieved **10× lower computation cost** than exhaustive and reachability-based methods with **near-optimal** task performance under obstacle and disturbance scenarios. [P3]

Selected Publications

- [P1] **Ruiyang Wang**, Hao-lun Hsu, David Hunt, Shaocheng Luo, Jiwoo Kim and Miroslav Pajic, “LLM-MCoX: Large Language Model Multi-robot Coordinated Exploration and Search”, Submitted to *International Conference on Robotics & Automation (ICRA)* 2026.
Video demonstrations available from https://ruiyangwangw.github.io/assets/ICRA_2026_LLM-MCoX_video.mp4
- [P2] **Ruiyang Wang**, Bowen He, Miroslav Pajic, “Neural Symbolic Deadlock Resolution in Multi-robot Systems”, Accepted to *Learning for Dynamics & Control (L4DC)* 2025.
Available from <https://proceedings.mlr.press/v283/wang25f.html>
- [P3] **Ruiyang Wang***, Hardik Parwana*, Dimitra Panagou, “Algorithms for Finding Compatible Constraints in Receding-Horizon Control of Dynamical Systems”, Accepted to *American Control Conference (ACC)* 2024.
Available from <https://ieeexplore.ieee.org/abstract/document/10644243>

Course Projects

ROB 498 Robot Learning for Planning and Control

January 2023 - April 2023

Pytorch, PyBullet, OpenAI Gym, VAE

- Built a latent-space controller for a 7-DOF **PANDA robotic arm** using VAE and SINDy.
- Achieved **20% fewer actions** per episode than baseline controllers, demonstrating improved control efficiency and stability.

ROB 599 Deep Learning for Robot Perception

January 2023 - April 2023

PoseCNN, PyTorch

- Reimplemented the **PoseCNN** network for 6D object pose estimation in **PyTorch**. Trained and evaluated the network on PROPS Pose Estimation Dataset and achieved a **5°5cm Accuracy** of **53.62%**.
- Investigated the effect of feed-forward layers in the network and improved the 5°5cm Accuracy to **64.14%** by optimizing feed-forward network depth.

ROB 550 BotLab

September 2022 - December 2022

*ROS2, C++, PID Control, Particle Filter SLAM, A**

- Built a fully autonomous maze-exploration robot with real-time localization and **A*** path planning.
- Integrated PID-based motor control and **particle filter localization** achieving robust navigation over different maze layouts.